Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C, such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. Let C and C' be two nonsingular curves such that X is birational to $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$. Then, there exists rational maps $\pi, \pi' : X \dashrightarrow C, C'$ and sections $\sigma, \sigma' : C, C' \to X$ such that $\pi \circ \sigma$ and $\pi' \circ \sigma'$ extend to the identity morphism on C and C', respectively. Then $\pi' \circ \sigma : C \to C'$ and $\pi \circ \sigma' : C' \to C$ are inverses to each other, so we have an isomorphism $C \cong C'$. □

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C. Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. If $\mathcal{E}$ is decomposable, then $\mathcal{E} = \mathcal{O} \oplus \mathcal{L}$, where $\mathcal{L} = \mathcal{O}_C(\mathfrak{e})$ for some divisor $\mathfrak{e} \in \operatorname{Div} C$. By (2.8.1), (2.11.1), and (2.11.2), the two natural projections $\mathcal{E} \to \mathcal{L}$ and $\mathcal{E} \to \mathcal{O}_C$ correspond to sections $C' \sim C_0$ and $C'' \sim C_0 - \mathfrak{e}f$, respectively, such that $C'.C'' = 0$. Hence, $C' \cap C'' = \emptyset$. Conversely, suppose there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$. The corresponding surjective morphisms $\mathcal{E} \to \mathcal{L}'$ and $\mathcal{E} \to \mathcal{L}''$ induce an $\mathcal{O}_C$-module homomorphism $\mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$. For every $P \in C$, the morphism of stalks $\mathcal{E}_P \to \mathcal{L}'_P \oplus \mathcal{L}''_P$ is an isomorphism because C' and C'' are disjoint, so $\mathcal{L}'_P \oplus \mathcal{L}''_P$ is generated by $\mathcal{E}_P$. Hence, $\mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$ is an isomorphism. □

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C, then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \ldots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

(b) Show that this is false for varieties of dimension $\geqslant 2$. In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

Proof.

(a) We proceed by induction on r. The base case $r = 1$ is trivial, so assume the statement for any $r > 1$. We can twist $\mathcal{E}$ by a very ample sheaf $\mathcal{L}$ so that $\mathcal{E}' = \mathcal{E} \otimes \mathcal{E}$ is generated by global sections. If we prove the statement for $\mathcal{E}'$ so we have a filtration $(\mathcal{E}'_i)$, then tensoring by $\mathcal{L}^{-1}$ gives a filtration $(\mathcal{E}_i = \mathcal{E}'_i \otimes \mathcal{L}^{-1})$ of $\mathcal{E}$ such that

$$\mathcal{E}_i/\mathcal{E}_{i-1} \cong (\mathcal{E}'_i/\mathcal{E}'_{i-1}) \otimes \mathcal{L}^{-1},$$

which is an invertible sheaf for each $i = 1, \ldots, r$. Hence, we can assume $\mathcal{E}$ is generated by global sections. The hypothesis of (II, Ex. 8.2) is satisfied for $\mathcal{E}$, so there exists a short exact

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{F}$ is also globally generated locally free of rank $r-1$. By the inductive hypothesis, this means we can find a filtration $(\mathcal{F}_i)$ of $\mathcal{F}$ such that each successive quotient is an invertible sheaf. Consider the preimage $(\mathcal{E}_j)$ of $(\mathcal{F}_i)$, where we relabel $j = i+1$. Each $\mathcal{E}_j$ is also locally free, $\mathcal{E}_1 = \operatorname{im}(\mathcal{O}_C)$ by exactness, and there is a canonical isomorphism $\mathcal{E}_j/\mathcal{E}_{j-1} \cong \mathcal{F}_{j-1}/\mathcal{F}_{j-2}$. Extending $(\mathcal{E}_j)$ with $\mathcal{E}_0 = 0$ gives a successive extension of sheaves.

(b) Suppose we have an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \Omega \longrightarrow \mathcal{M} \longrightarrow 0,$$

where $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on $\mathbb{P}^2$. Then $H^1(\mathcal{L}), H^1(\mathcal{M}) = 0$, so by the long exact of cohomology, $H^1(\Omega) = 0$. But by (III, Ex. 7.2), $H^1(\Omega) = k$, a contradiction.

□

4. Let C be a curve of genus g, and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

(a) Show that $r = 0$ and $r \geqslant g + 1$ are always possible.

(b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Proof. Fix a section $C_0 = C \times P_0 \subset X$, where P_0 is a point of $\mathbb{P}^1$. The ruled surface $X = C \times \mathbb{P}^1$ corresponds to the projective bundle of $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C$. The invariant of X is $e = 0$, which means $C_0^2 = -e = 0$. By (2.9), if D is any section of X corresponding to a surjection $\mathcal{E} \to \mathcal{O}_C(\mathfrak{d})$ for some divisor $\mathfrak{d}$ of C with $r = \deg \mathfrak{d}$, then $r = C_0.D$ and $D \sim C_0 + \mathfrak{d}f$. Hence, $s = 2r$.

(a) There is a one-to-one correspondence between sections $C \to D \subset X$ and finite morphism $f : C \to \mathbb{P}^1$. In particular, a section $D \subset C \times \mathbb{P}^1$ is the graph of a morphism $f : C \to \mathbb{P}^1$, and its self-intersection is $D^2 = 2 \cdot \deg f$. By (IV, Ex. 1.6), $g + 1$ is an upperbound on the minimum of $\deg f$. Hence, $r = 0$ and $r \geqslant g + 1$ is always possible.

(b) If $g > 0$, then there is no morphism $C \to \mathbb{P}^1$ of degree 1. If C is a hyperelliptic curve, then it has a g^1_2, so $r = 2, 3$ is possible. Otherwise, nonhyperelliptic curves of genus 3 always have a g^1_3 (IV, 5.5.2), so $r = 3$ is always possible.

□

5. *Values of* e. Let C be a curve of genus $g \geqslant 1$.

(a) Show that for each $0 \leqslant e \leqslant 2g - 2$ there is a ruled suface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in H^1(X, \mathcal{O}_X(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_X(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(X, \mathcal{O}_X(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$.

(c) Now show that if $-g \leqslant e < 0$, there exists a ruled surface X over C with invariant e.

(d) For $g = 2$, show that $e \geqslant -2$ is also necessary for the existence of X.

Proof.

(a) Let $\mathfrak{e}$ be a divisor of degree $-e$ on C, and let $\mathfrak{k}$ is a canonical divisor of C. We need to first construct an indecomposable extension $\mathcal{E}$ of $\mathcal{O}_C(\mathfrak{e})$. By (III, Ex. 6.1), an extension of vector bundles

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(\mathfrak{e}) \longrightarrow 0$$

corresponds to an element in $\operatorname{Ext}^1(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C)$. In this case, $\operatorname{Ext}^1(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C) \cong H^1(C, \mathcal{O}_C(-\mathfrak{e}))$ (III, 6.3) and (III, 6.7). This is dual to $H^0(C, \mathcal{O}_C(\mathfrak{k} + \mathfrak{e}))$, so it is enough to find $\mathfrak{e}$ such that $H^0(C, \mathcal{O}_C(\mathfrak{k} + \mathfrak{e})) \neq 0$. The image of $C \to |\mathfrak{k}| \cong \mathbb{P}^{g-1}$ is a curve of degree either $g - 1$ or $2g - 2$. If

$-\mathfrak{e}$ is effective, then $H^0(C, \mathcal{O}_C(\mathfrak{k}+\mathfrak{e}))$ is the space of all hyperplane sections in $\mathbb{P}^{g-1}$ containing the support $\mathfrak{e}$. A general hyperplane section of $C \subset \mathbb{P}^{g-1}$ consists of $2g-2$ points counted with multiplicity. Hence, for each $0 \leqslant e \leqslant 2g-2$, there is a divisor $\mathfrak{e}$ of degree $-e$ such that $H^0(C, \mathcal{O}_C(\mathfrak{k}+\mathfrak{e})) \neq 0$, so there exists indecomposable extensions of $\mathcal{O}_C(\mathfrak{e})$ by $\mathcal{O}_C$.

Now, let $\mathcal{E}$ be an extension of $\mathcal{O}_C(\mathfrak{e})$ defined by a nonzero element in $\operatorname{Ext}^1(\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C)$. It remains to show $\mathcal{E}$ is normalized. Clearly $H^0(C, \mathcal{E}) \neq 0$ by construction. If $\mathcal{M}$ is any invertible sheaf, then we have an exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}_C(\mathfrak{e}) \otimes \mathcal{M} \longrightarrow 0.$$

If $\deg \mathcal{M} < 0$, then we have $H^0(C, \mathcal{M}) = 0$ and $H^0(C, \mathcal{O}_C(\mathfrak{e}) \otimes \mathcal{M}) = 0$, and therefore also $H^0(C, \mathcal{E} \otimes \mathcal{M}) = 0$. Hence, for each $0 \leqslant e \leqslant 2g-2$, there is a ruled surface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Suppose $\mathcal{E}$ is normalized, and let $\mathcal{M} = \mathcal{O}_C(E)$ be any invertible sheaf associated to a divisor E of degree $d-1$.

(c)

(d)

□

6. (*) Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

Proof. We proceed by induction on the rank. Let $\mathcal{E}$ be a locally sheaf of rank $r > 0$, and assume the statement for $< r$. First, note that the definition of normalization and the statement of (2.8) holds for locally free sheaves of arbitrary rank on any curve.

Proposition 0.1. *If $\mathcal{E}$ is a locally free sheaf of finite rank on a curve C, then there exists an invertible sheaf $\mathcal{L}$ on $\mathbb{P}^1$ such that if $\mathcal{E}' = \mathcal{E} \otimes \mathcal{L}$, then $H^0(C, \mathcal{E}') \neq 0$, and for all invertible sheaves* $\deg \mathcal{M} < 0$, *we have* $H^0(C, \mathcal{E}' \otimes \mathcal{M}) = 0$.

An invertible sheaf of positive degree on C is ample (IV, 3.3), so it is possible to make $H^0(C, \mathcal{E}') \neq 0$ by taking $\deg \mathcal{L}$ large enough. On the other hand, by (II, Ex. 8.2), since an invertible sheaf of negative degree can have no global sections, we see that $H^0(C, \mathcal{E}') = 0$ for $\deg \mathcal{L}$ sufficiently negative. So we achieve our result by taking an $\mathcal{L}$ of least degree such that $H^0(C, \mathcal{E}') \neq 0$.

After multiplying by a suitable choice of an invertible sheaf, we can assume $\mathcal{E}$ is generated by global sections. Picking a section in $H^0(\mathbb{P}^1, \mathcal{E})$ is equivalent to defining an injection $\mathcal{O} \to \mathcal{E}$, so pick $\mathcal{O} \to \mathcal{E}$ of maximal degree. I claim the quotient $\mathcal{F} = \mathcal{E}/\mathcal{O}$ is locally free of rank $r-1$. If not, let $\mathcal{D} \subset \mathcal{E}$ be the inverse image of the torsion subsheaf of $\mathcal{F}$ by the map $\mathcal{E} \to \mathcal{F} \to 0$. In that case, $\mathcal{D}$ is locally free of rank $< r$, so we can apply the inductive hypothesis. Furthermore, $\mathcal{O}_C \subset \mathcal{T}$

Now, consider the following short exact sequence of $\mathcal{O}_{\mathbb{P}^1}$-modules

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0.$$

By the inductive step, we can write $\mathcal{F} = \bigoplus \mathcal{O}_{\mathbb{P}^1}(n_i)$ for some $n_i \in \mathbb{Z}$. □

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C, and that no two are linearly equivalent.

Proof. If C_0 is a section of X such $C_0^2 = 1$, then C_0 corresponds to a divisor $\mathfrak{d}$ of degree one on C, which is one-to-one correspondence with the closed points of C. In the other direction, fix a point $P \in C$ so that $\mathcal{E}$ is the non-trivial extension of $\mathcal{O}$ and $\mathcal{O}(P)$ (2.11.6). If Q is any point in C, including the case $Q = P$, then $\operatorname{Ext}^1(\mathcal{O}(Q), \mathcal{O}(P - Q))$ □

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \to \mathcal{F} \to 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F}) / \operatorname{rank} \mathcal{F} > (\deg \mathcal{E}) / \operatorname{rank} \mathcal{E}$$

Replacing $>$ by $\geqslant$ defines *semistable.*

(a) A decomposable $\mathcal{E}$ is never stable.

(b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E}.0$ (respectively, $\geqslant 0$).

(c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not seistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leqslant 2g-2$, (2) an element $\mathcal{L} \in \operatorname{Pic} C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(X, \mathcal{L}^{-1})$, deterimned up to a nonzero scalar multiple.

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geqslant 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.

10. For any $n > e \geqslant 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geqslant 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geqslant 4$, there exists a nonhyperelliptic curve which has a g_3^1.

11. Let X be a ruled surface over the curve C, defined by a normalized bundle $\mathcal{E}$, and let $\mathfrak{e}$ be the divisor on C for which $\mathcal{O}_C(\mathfrak{e}) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C.

(a) If $|\mathfrak{b}|$ and $|\mathfrak{b}+\mathfrak{e}|$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no points.

(b) If $\mathfrak{b}$ and $\mathfrak{b}+\mathfrak{e}$ are very ample on C, and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b}+\mathfrak{e}-P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.

12. Let X be a ruled surface with invariant $\mathfrak{e}$ over an elliptic curve C, and let $\mathfrak{b}$ be a divisor on C.

(a) If $\deg \mathfrak{b} \geqslant \mathfrak{e}+2$, then there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.

(b) The linear system $|C_0 + \mathfrak{b}f|$ is very ample if and only if $\deg \mathfrak{b} \geqslant \mathfrak{e}+3$.

13. For every $e \geqslant -1$ and $n \geqslant e+3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d-1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.

14. Let X be a ruled surface over a curve of genus g, with invariant $e < 0$, and assume that $\operatorname{char} k = p > 0$ and $g \geqslant 2$.

(a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0.f$, then either $a = 1$, $b \geqslant 0$, or $2 \leqslant a \leqslant p-1$, $b \geqslant \frac{1}{2}ae$, or $a \geqslant p$, $b \geqslant \frac{1}{2}ae + 1 - g$.

(b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.

15. *Funny behaviour in characteristic* p. Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3.

(a) Show that the action of the k-linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0.

(b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ such that $f^*\xi = 0$ in $H^1(C, \mathcal{O}_C(-3P))$.

(c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let X be the corresponding ruled surface over C. Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi : Y \to C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypothesis of (2.21b), but is not ample.

16. Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

Proof. Let A be a Dedekind domain finitely generated over k such that $C \cong \operatorname{Spec} A$. The category of coherent $\mathcal{O}_C$-modules is equivalent to the category of finitely A-modules. Under this identification, $K(X)$ is naturally isomorphic to the Grothendieck group of flat finitely generated A-modules $K_0(A)$. The structure theorem of modules over Dedekind domains states $K_0(A) \cong \mathbb{Z} \oplus \mathrm{Cl}(A)$, where if M is any flat finitely generated A-module, then $M \cong A^r \oplus I$, where K is the fraction field of A, $r + 1 = \dim_K M \otimes_A K$ is the rank of M, and I is some fractional ideal in $\mathrm{Cl}(A)$. If $N \cong A^s \oplus J$ is another A-module, then $M \cong N$ if and only if $r = s$ and $I \cong J$ in $\mathrm{Cl}(A)$. $\square$

17. (a) Let $\varphi : \mathbb{P}^1_k \to \mathbb{P}^3_k$ be the 3-uple embedding. Let $\mathcal{J}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ. Then $\mathcal{J}/\mathcal{J}^2$ is a locally free sheaf of rank 2 on C, so $\varphi^*(\mathcal{J}/\mathcal{J}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{J}/\mathcal{J}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m.

(b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \to \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.